



Data Quality Assessment Report

Booked Orders.xlsx

Sheet: Booked Orders (New - YTD)

80.9

Overall Quality Score / 100

Good

Processing Date: 2026-08-05 12:49:26

Engine Version: Phase 1

Execution Time: 21.17s

Generated By: Data Quality Engine (Phase 1)

Executive Summary

Dataset:	Booked Orders.xlsx
Rows / Columns:	101,352 rows x 23 columns
Checks Executed:	8
Overall Quality Score:	80.9 / 100 (Good)

Critical Findings

- Outliers: 10 column(s) affected, 133199 issue(s) found
- Validity: 6 column(s) affected, 184843 issue(s) found
- PII detected in 12 column(s) -- masking required before sharing

Positive Findings

- Missing Values: no issues found
- Duplicates: no issues found
- Type Mismatch: no issues found
- Schema Quality: no issues found

Data Readiness

Ready with Minor Cleaning

Business Impact & Risk Summary

Outliers

Critical

Business Impact: May distort averages, totals, and any ML model trained on this data.

Consistency

Medium

Business Impact: Grouping and aggregation become unreliable (e.g. 'Paid' vs 'paid' split apart).

Validity

High

Business Impact: Values that look present may still be unusable or logically wrong.

Freshness

Medium

Business Impact: Decisions may be based on stale data without anyone noticing.

PII Exposure

Critical

Business Impact: Privacy and compliance risk if this data is shared unmasked.

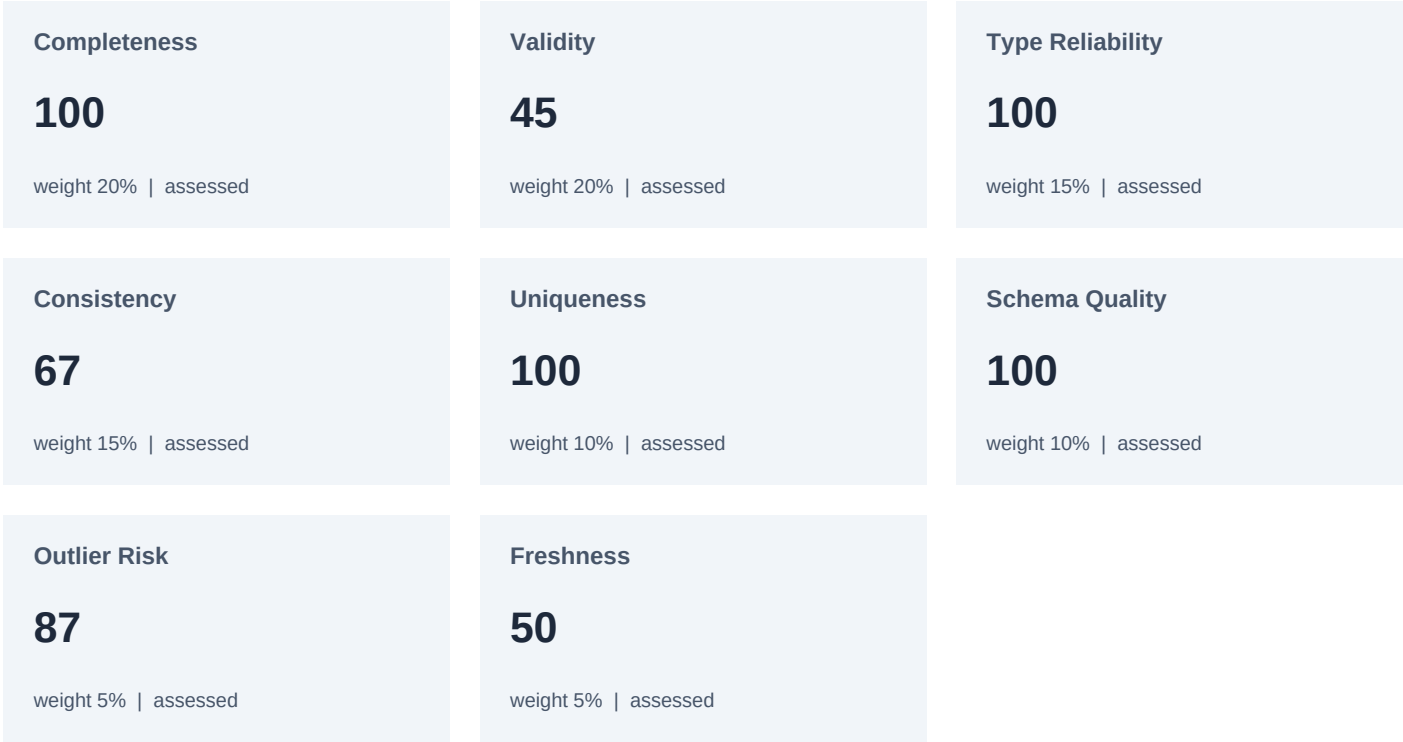
Dataset Overview

Rows:	101,352
Columns:	23
Header Row:	1
Processing Time:	21.17s

Column Classification

Measurement Columns:	10
Identifier Columns:	5
PII Columns:	0
Free-Text Columns:	0
Categorical Columns:	4
Date Columns:	4

Data Quality Dashboard



Dimension Scores at a Glance



Privacy Risk (separate from score above)



Types found: CARD, CNIC, IBAN, IP_ADDRESS, MOBILE, PHONE, SSN

Missing Values

None

Columns Checked:	23
Columns with Issues:	0
Total Issues Found:	0

Business Impact

May produce inaccurate analytics and unreliable reporting.

Recommendation

Improve data collection at source; consider a required-field rule for critical columns.

Duplicates

None

Columns Checked:	1
Columns with Issues:	0
Total Issues Found:	0

Business Impact

Can create duplicate customers/orders and inflate revenue or count metrics.

Recommendation

Add a uniqueness constraint on the business key at the source system.

Type Mismatch

None

Columns Checked:	23
Columns with Issues:	0
Total Issues Found:	0

Business Impact

Charting, grouping, and trend analysis may silently break or mislead.

Recommendation

Standardize the column's expected type at data entry.

Outliers

Critical

Columns Checked:	13
Columns with Issues:	10
Total Issues Found:	133199
Affected Columns:	Cost, Cost_1, Exchange Rate, GBP Amt-tax, Line Status Flag, Margin, Margin%, Margin_1, Quantity Ordered STU, Transaction Currency Amt - tax

Business Impact

May distort averages, totals, and any ML model trained on this data.

Recommendation

Manually review flagged values before using this column in analysis or ML.

Sample Findings

- Quantity Ordered STU: 13064 issue(s)
- GBP Amt-tax: 14323 issue(s) note=high concentration on one value -- IQR bounds may be unreliable for this distribution
- Cost: 13953 issue(s)
- Margin: 22292 issue(s)
- Margin%: 6160 issue(s) note=high concentration on one value -- IQR bounds may be unreliable for this distribution

Schema Quality

None

Columns Checked:	23
Columns with Issues:	0
Total Issues Found:	0

Business Impact

Downstream tools and analysts may misread or skip unclear columns.

Recommendation

Rename unclear/duplicate/blank columns before this file is reused.

Consistency

Medium

Columns Checked:	23
Columns with Issues:	3
Total Issues Found:	1560
Affected Columns:	Cust Order Ref Cust Ord Ref, Manufacturers Ref, Prod Desc

Business Impact

Grouping and aggregation become unreliable (e.g. 'Paid' vs 'paid' split apart).

Recommendation

Standardize values at entry (dropdowns/lookups instead of free text).

Sample Findings

- Cust Order Ref Cust Ord Ref: 1490 issue(s) examples=[{'normalized': 'texecom dd', 'canonical': 'TEXECOM DD', 'variants': {'TEXECOM DD': 147, 'texecom dd': 3, 'Texecom DD': 9}}, {'normalized': 'texecom urgent', 'canonical': 'TEXECOM URGENT', 'variants': {'TEXECOM URGENT': 1, 'texecom urgent': 3, 'Texecom URGENT': 9}}]
- Manufacturers Ref: 6 issue(s) examples=[{'normalized': 'adum1401arwz', 'canonical': 'ADUM1401ARWZ', 'variants': {'ADUM1401ARWZ': 1, 'ADuM1401ARWZ': 1}}, {'normalized': 'tool-0b101/01/3/06/i8', 'canonical': 'TOOL-0B101/01/3/06/i8', 'variants': {'TOOL-0B101/01/3/06/i8': 1, 'tool-0b101/01/3/06/i8': 1}}]
- Prod Desc: 64 issue(s) examples=[{'normalized': 'res 10r 0603 1%', 'canonical': 'Res 10R 0603 1%', 'variants': {'Res 10R 0603 1%': 8, 'RES 10R 0603 1%': 6}}, {'normalized': 'led 2012 red', 'canonical': 'Led 2012 Red', 'variants': {'LED 2012 Red': 1, 'led 2012 red': 1}}]

Validity

High

Columns Checked:	14
Columns with Issues:	6
Total Issues Found:	184843
Affected Columns:	Cost, Date Expected Del., GBP Amt-tax, Line Status Flag, Margin%, Sales Order Line Expected Del. Date

Business Impact

Values that look present may still be unusable or logically wrong.

Recommendation

Fix values that fail format/logic rules at the source before further use.

Sample Findings

- Line Status Flag: 101352 issue(s)
- Sales Order Line Expected Del. Date: 4 issue(s) rule=invalid_or_implausible_date
- Date Expected Del.: 2 issue(s) rule=invalid_or_implausible_date
- GBP Amt-tax: 31105 issue(s) rule=suspicious_zero_value
- Cost: 17421 issue(s) rule=suspicious_zero_value

Freshness

Medium

Columns Checked:	23
Columns with Issues:	2
Total Issues Found:	2
Affected Columns:	Order Date, SO Date Created

Business Impact

Decisions may be based on stale data without anyone noticing.

Recommendation

Confirm this data is refreshed on the expected schedule.

Sample Findings

- SO Date Created: 1 issue(s)
- Order Date: 1 issue(s)

Column Quality Matrix

Column	Role	Score	Issues	Severity	Recommendation
GBP Amt-tax	measurement	71	45428	High	Manually review flagged values before using this column
Cost	measurement	71	31374	High	Manually review flagged values before using this column
Margin%	measurement	71	41119	High	Manually review flagged values before using this column
Line Status Flag	measurement	71	108920	High	Review flagged values.
SO Date Created	date	83	1	Medium	Confirm this data is refreshed on the expected schedule
Order Date	date	83	1	Medium	Confirm this data is refreshed on the expected schedule
Sales Order Line Expected D	date	83	4	Medium	Fix values that fail format/logic rules at the source b
Date Expected Del.	date	83	2	Medium	Fix values that fail format/logic rules at the source b
Cust Order Ref Cust Ord Ref	identifier	83	1490	Medium	Standardize values at entry (dropdowns/lookups instead
Manufacturers Ref	identifier	83	6	Medium	Standardize values at entry (dropdowns/lookups instead
Prod Desc	categorical	83	64	Medium	Standardize values at entry (dropdowns/lookups instead
Margin	measurement	83	22292	Medium	Manually review flagged values before using this column
Exchange Rate	measurement	83	5329	Medium	Manually review flagged values before using this column
Transaction Currency Amt - t	measurement	83	14418	Medium	Manually review flagged values before using this column
Cost_1	measurement	83	13957	Medium	Manually review flagged values before using this column
Margin_1	measurement	83	22135	Medium	Manually review flagged values before using this column
Quantity Ordered STU	measurement	89	13064	Medium	Manually review flagged values before using this column
Order Number	identifier	100	0	None	No action needed.
Sold-to Name	categorical	100	0	None	No action needed.
Site Code	identifier	100	0	None	No action needed.
Ext Rep Name	categorical	100	0	None	No action needed.
Int Rep Name	categorical	100	0	None	No action needed.
Currency Code	identifier	100	0	None	No action needed.

Top Critical Issues

1. Outliers -- Cost

Critical

Impact: May distort averages, totals, and any ML model trained on this data.

Action: Manually review flagged values before using this column in analysis or ML.

2. Outliers -- Cost_1

Critical

Impact: May distort averages, totals, and any ML model trained on this data.

Action: Manually review flagged values before using this column in analysis or ML.

3. Outliers -- Exchange Rate

Critical

Impact: May distort averages, totals, and any ML model trained on this data.

Action: Manually review flagged values before using this column in analysis or ML.

4. Outliers -- GBP Amt-tax

Critical

Impact: May distort averages, totals, and any ML model trained on this data.

Action: Manually review flagged values before using this column in analysis or ML.

5. Outliers -- Line Status Flag

Critical

Impact: May distort averages, totals, and any ML model trained on this data.

Action: Manually review flagged values before using this column in analysis or ML.

6. Outliers -- Margin

Critical

Impact: May distort averages, totals, and any ML model trained on this data.

Action: Manually review flagged values before using this column in analysis or ML.

7. Outliers -- Margin%

Critical

Impact: May distort averages, totals, and any ML model trained on this data.

Action: Manually review flagged values before using this column in analysis or ML.

8. Outliers -- Margin_1

Critical

Impact: May distort averages, totals, and any ML model trained on this data.

Action: Manually review flagged values before using this column in analysis or ML.

9. Outliers -- Quantity Ordered STU

Critical

Impact: May distort averages, totals, and any ML model trained on this data.

Action: Manually review flagged values before using this column in analysis or ML.

10. Outliers -- Transaction Currency Amt - tax**Critical**

Impact: May distort averages, totals, and any ML model trained on this data.

Action: Manually review flagged values before using this column in analysis or ML.

Appendix

Execution Time: 21.17s
Engine Version: Phase 1

Scoring Formula

Each dimension score = $100 \times (\text{passed checks} / \text{total non-error checks})$ for that dimension. Composite score = weighted average across scorable dimensions only (weights re-normalized when a dimension has no results). Privacy Risk is calculated and reported separately and is never subtracted from the composite score.

Quality Dimensions

- Completeness (weight: 20%)
- Validity (weight: 20%)
- Type Reliability (weight: 15%)
- Consistency (weight: 15%)
- Uniqueness (weight: 10%)
- Schema Quality (weight: 10%)
- Outlier Risk (weight: 5%)
- Freshness (weight: 5%)